Universal Object-Level Adversarial Attack in Hyperspectral Image Classification

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***Abstract*— The vulnerability of deep neural networks (DNNs) has garnered significant attention. Various advanced adversarial attack methods have been proposed. However, these methods exhibit higher attack performance on three-band natural images while struggling to handle high-dimensional attacks in terms of attack transferability and robustness. Hyperspectral images (HSIs), unlike natural images, possess high-dimensional and redundant spectral information. On the one hand, different classification models focus on distinct discriminative spectral bands, leading to poor transferability. On the other hand, most existing attack methods are implemented at the pixel level, making them less resilient to image-processing-based defenses. In this article, we address the improvement of transferability and robustness in high-dimensional attacks and introduce a universal object-level adversarial attack method in HSI classification. We found that perturbations with higher similarity in a local region can decrease the sensitivity of adversarial attacks to various discriminative spectral patterns and enhance resistance to image-processing-based defenses. Consequently, we construct spatial and spectral oversegmented templates by utilizing the local smooth properties of HSIs, aiming to promote similarity among perturbations within a local region. Extensive experiments conducted on two real HSI datasets validate that our method enhances the attack transferability and robustness of several existing attack methods. By incorporating the object-level adversarial attack with the baseline fast gradient sign method (FGSM), momentum iterative FGSM (MI-FGSM), and variance tuning MI-FGSM (VMI-FGSM), the average transferability success rate of the proposed method has increased by 7.38% on the PaviaU dataset and 9.30% on the HoustonU 2018 dataset than the baselines. Meanwhile, the proposed method outperforms the baselines by an average of 6.19% on the PaviaU dataset and 10.05% on the HoustonU 2018 dataset in attacking image-processing-based defense models. The code is available at https://github.com/AAAA-CS/SS_FGSM_HyperspectralAdversarialAttack.**

***Index Terms*— Adversarial attack, adversarial defense, hyperspectral image (HSI) classification.**

I. Introduction

REMOTE-SENSING (RS) image classification plays a crucial role in comprehending land-cover objects [1], [2], [3]. Hyperspectral images (HSIs) are capable of capturing not only the 2-D spatial characteristics of land covers, but also fine spectral information, enabling more accurate recognition of land-cover objects. Deep-learning technology has emerged as a prominent research approach in the field of HSI classification. Deep neural networks (DNNs) are employed to extract rich and complementary spectral–spatial features, thereby enhancing the classification performance [4], [5], [6].

Nevertheless, the vulnerability of DNNs to adversarial examples has garnered significant attention [7], [8]. By introducing imperceptible perturbations to benign examples, DNNs can yield incorrect predictions. This issue has stimulated research on adversarial attacks and defenses, aiming to comprehend and assess the associated risks and enhance the robustness of DNNs. However, despite the remarkable success of most adversarial attack methods on natural images, their application for generating adversarial examples in the high-dimensional space of HSIs encounters two challenges.

1) *Poor Transferability in High-Dimensional Image Data:* Adversarial attack methods can be categorized into white-box and black-box attacks, with the latter being more challenging in real-world scenarios. In particular, transfer-based black-box attacks have attracted significant interest as they aim to exploit the transferability of adversarial examples crafted by white-box attack methods [9]. However, existing studies primarily focus on adversarial attacks for one- or three-band images, while few works address adversarial attacks on high-dimensional image data. In the case of HSIs, which have an additional spectral dimension and strong spectral redundancy compared to natural images, we observed that different classification models exhibit similar spatial attention information, but significant differences in discriminative spectral bands. More detailed analysis can be found in Section III-A. Existing attack methods lack consideration for spectral-dimension differences, leading to a decrease in high-dimensional attack transferability. To enhance the transferability of HSI adversarial examples, it becomes crucial to reduce the sensitivity of attack methods to different discriminative spectral patterns.
2) *Poor Robustness to Image-Processing-Based Defenses:* Many adversarial attack methods traditionally operate at the pixel level, where perturbations are applied independently to each pixel. However, pixel-level perturbations are often not robust against image-processing-based defenses [9]. In HSI classification tasks, HSIs usually suffer from image preprocessing technologies to

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remove noise, such as spectral and spatial filtering, as well as spatial resizing. Perturbations with local smooth characteristics can more effectively against image-processing-based defenses.

To address the aforementioned challenges in adversarial attacks on HSI classification networks, we propose a novel universal object-level adversarial attack method by promoting local smoothness in spatial and spectral perturbations. On the one hand, compared to pixel-level perturbations, locally smoothed perturbations prevent adversarial examples from overfitting white-box models, and locally smoothed spectral-dimension perturbations reduce sensitivity to discriminative patterns across different classification models, enhancing the transferability of HSI adversarial examples from white-box to black-box models. On the other hand, the spatial and spectral information of HSIs is locally smooth, and therefore, more smoothed perturbations can preserve more powerful attacks to image-processing-based defenses, making the perturbations harder to filter out.

To achieve locally smooth high-dimensional perturbations, we draw inspiration from [9] and utilize the superpixel segmentation method [10] to create an oversegmented spatial template; additionally, the k-means clustering method [11] is employed to produce an oversegmented spectral template. Subsequently, the local smoothness of spatial and spectral perturbations is enhanced under the constraints of spatial and spectral templates. Our approach integrates the spatial and spectral attacks in a unified framework, and the spatial and spectral perturbations are optimized iteratively. To the best of our knowledge, this work represents the first exploration of object-level attacks on high-dimensional image data, specifically targeting HSIs. By focusing on object-level attacks and leveraging the advantages of smooth perturbations, our method aims to enhance the robustness and transferability of adversarial examples in HSI classification networks.

The main contributions are summarized as follows.

1) We present a universal adversarial attack approach specifically designed for HSI classification tasks. The proposed object-level attack method can be easily integrated into existing gradient-based attack methods to enhance their transferability and robustness.
2) We analyze the differences in discriminative spectral patterns of various HSI classification networks and generate locally smoothed spectral perturbations to reduce the sensitivity of perturbations to discriminative spectral patterns.
3) We propose a high-dimensional spectral–spatial object-level perturbation generation framework to enhance the robustness of perturbations against image-processing-based defense methods.
4) We evaluate the effectiveness of the proposed method by integrating the spectral–spatial object-level perturbation generation framework with three gradient-based attack methods. Through extensive experiments conducted on two real HSI datasets, we compare the performance of our method with state-of-the-art approaches in terms of transferability and robustness. The results and analyses clearly demonstrate that our proposed method outperforms existing methods in attack transferability and robustness.

II. Background

A. Adversarial Attacks

Recently, several adversarial attack methods have been proposed to craft powerful adversarial examples. The classical gradient-based attack method is the fast gradient sign method (FGSM) [12], which generates an adversarial example by maximizing the prediction of the networks and true label, and then the FGSM is extended iteratively to further improve the attack performance [13]. After that, most gradient-based attack methods were implemented iteratively [14], [15]. More recently, black-box attacks, which do not access the gradient information of the target model, have attracted more attention. To improve the power of black-box attacks, several methods were proposed to improve the transferability of the adversarial example crafted on the white-box model. The classical transfer-based attack is the momentum iterative FGSM (MI-FGSM) [13], which introduces the moment term into the attack process to craft more transferable adversarial examples. Based on MI-FGSM, more advanced momentum techniques were proposed, such as Nesterov iterative FGSM (NI-FGSM) [16], variance tuning MI-FGSM (VMI-FGSM) [17], and adaptive momentum variance iterative FGSM (AMVI-FGSM) [18]. Another way is to enhance the transferability by adopting diverse input transformations, such as the diverse input method (DIM) [19], the translation-invariant method (TIM) [20], and the scale-invariant method (SIM) [16]. However, these methods focus more on improving the attack transferability on one- or three-band natural images and are not robust to image-processing-based defenses.

Among the existing attack methods, the most related work to ours is superpixel-guided attentional version I-FGSM (SAI-FGSM) [9], which proposed a superpixel-guided adversarial attack method. Nevertheless, there are several differences between SAI-FGSM and our method, that is, different attack targets and different optimization strategies. First, SAI-FGSM aims to generate a three-band adversarial example by utilizing the spatial local smooth property, while our method generates HSI adversarial examples from two different spatial and spectral views. Second, the SAI-FGSM strictly constrains the perturbations within each superpixel to be the same, resulting in a decrease in attack success rate on some models, while our method forces the perturbations within a high-dimensional object region to be similar in an iterative manner, which can significantly improve the attack success rate while maintaining attack robustness.

B. Adversarial Defense

The existing defense techniques can be categorized into two main groups: adversarial training [21], [22], [23] and preprocessing-based defense methods [24], [25]. Adversarial training methods utilize generated examples to improve the robustness of the networks. General example generation methods, such as data augmentation [26] and generative adversarial

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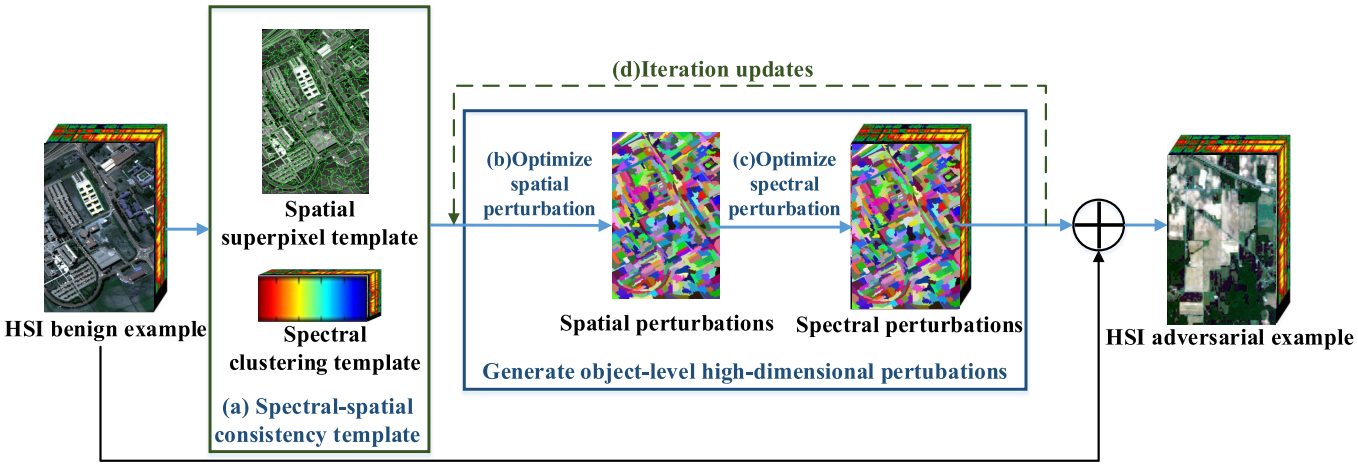


Fig. 1. Architecture of universal object-level adversarial attack in HSI classification. (a) Generate spatial and spectral templates via superpixel segmentation and k-means clustering. (b) Optimize spatial perturbation based on the spatial template. (c) Optimize spectral perturbations based on the spectral template. (d) Update spatial and spectral perturbations iteratively.

networks (GANs)-based example generation method [27], [28], can enhance the robustness of networks to a certain extent, however, there is no guarantee that the distribution of generated examples is close to the classification boundary. As a result, defense models based on general example generation methods may be insufficient against carefully crafted adversarial examples. Therefore, the adversarial training method aims to defend against attacks by integrating carefully crafted adversarial examples into the training process [29]. However, the general adversarial training methods have high computational costs. The preprocessing-based defense methods focus on removing the perturbations before feeding the adversarial examples into the target model, which include the GAN-based defense method and image-processing-based defense method. The typical GAN-based defense methods are adversarial perturbation erasing GAN (APE-GAN) [30], Defense-GAN [31], and attack-invariant attention feature (AIAF)-Defense [32]. The typical image-processing-based defense methods involve operations such as denoising [33], [34] and resizing [35] to remove perturbations. HSIs usually perform image preprocessing in spatial and spectral dimensions before classification. Therefore, in this study, image-processing-based defense methods are utilized to assess the adversarial robustness of attacks.

C. Adversarial Attacks and Defense in Remote Sensing

The vulnerability of DNNs has also attracted the interest of RS researchers. Xu et al. [36] and Chen et al. [37] investigated the performance of classical attack methods in remote-sensing scene classification tasks. Li et al. [38] and Du et al. [39] analyzed the sensitivity of attacks on SAR image classification and recognition tasks. Cheng et al. [40] proposed an adaptive adversarial example generative model via adversarial generative networks. These works mainly focus on adversarial attacks and adversarial robustness on three-band optical remote-sensing images and SAR images. Although the work in [41] investigated the vulnerability of HSI classification networks to adversarial examples and proposed a defense strategy by modifying the structure of the networks, the attack transferability in HSI classification networks was not studied.

III. Approach

In this section, we first elaborate on the motivation of this article, then describe the construction of spatial and spectral templates, and illustrate the attack process with spectral–spatial consistency constraints. The architecture of our method is shown in Fig. 1.

A. Motivation

Consider a benign HSI example $x \in X \subset R^{C\times H\times W}$, where H and W are the height and width, respectively, and C is the number of spectral bands. Consider y as the ground-truth label and f as the target model. We can find an adversarial perturbation $\delta \subset R^{C\times H\times W}$ in the x-neighbor to generate an adversarial example $x^{\text{adv}} \in X \subset R^{C\times H\times W}$, by optimizing

$$x^{\text{adv}} = \arg\max_{\|\delta\|\le\varepsilon} J(f(x+\delta), y) \tag{1}$$

where $J(\cdot)$ is the cross-entropy loss function and ε is the maximum perturbation budget. The existing works mainly focus on optimizing the three-band perturbations in a pixel-level manner and ignore the essential characteristics of the image.

Different from natural images with three bands, HSIs have high-dimensional and redundant data characteristics; therefore, different classification models may focus on different discriminative spectral bands, resulting in reduced transferability. We use Grad-CAM [42] to compare the discriminative spatial and spectral patterns among the five different classification models. The spatial attention map is defined in

$$L^{(\text{spa})} = \text{ReLU}\left(\sum_c w^{\text{spa},c} x^c\right) \tag{2}$$

where $w^{\text{spa},c} = (1/(H\times W))\sum_H\sum_W \partial f(x)/\partial x^c$, x^c denotes the cth band of HSIs and $w^{\text{spa},c}$ denotes the importance weight

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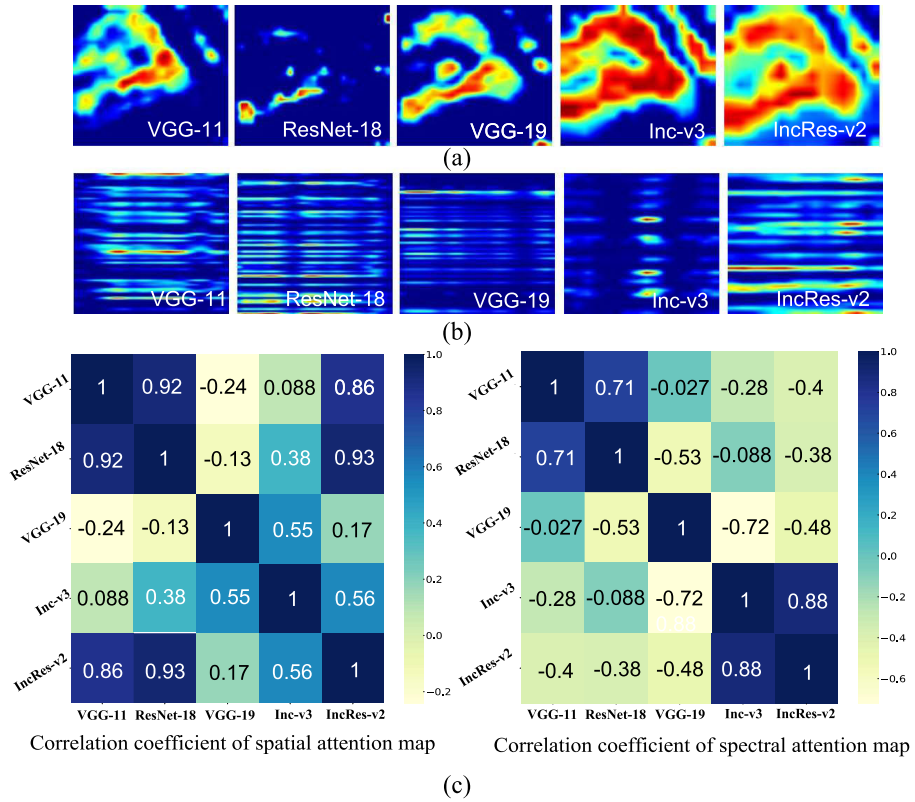


Fig. 2. Relation between spatial and spectral attention maps via Grad-CAM. (a) Spatial attention map. (b) Spectral attention map. (c) Relations between different classification models in high-attention regions (values > 0.5).

of each band. Similarly, the spectral attention map is defined in

$$L^{(\text{spe})} = \text{ReLU}\left(\sum_H w^{\text{spe},h} x^h + \sum_W w^{\text{spe},w} x^w\right) \tag{3}$$

where $w^{\text{spe},h} = (1/(W \times C)) \sum_W \sum_C \partial f(x)/\partial x^h$, $w^{\text{spe},w} = (1/(H \times C)) \sum_H \sum_C \partial f(x)/\partial x^w$, x^h and x^w denote the spectral cross section of HSIs along the height and width, respectively, and $w^{\text{spe},h}$ and $w^{\text{spe},w}$ denote the importance weights of each cross section, respectively. Examples of the spatial and spectral attention maps are visualized in Fig. 2(a) and (b), respectively. Compared to spatial attention maps, spectral attention maps have more significant differences. Therefore, it is difficult to obtain high-transferable HSI adversarial examples by considering only the spatial neighborhood information. Furthermore, Fig. 2(c) shows the relations between different classification models on spatial and spectral attention maps, respectively. The presented correlation coefficients further verify our assumption.

In addition, HSIs have local smooth properties in both spatial and spectral dimensions; hence, the pixel-level perturbations in the smooth regions are easily removed by image-processing-based defenses, such as spectral filtering and spatial resizing.

Based on the above considerations, we attempt to design a unified framework to enhance attack transferability and robustness by obtaining smoother high-dimensional spectral–spatial perturbations. Locally smoothed perturbations can reduce the sensitivity of attack methods on discriminative spectral patterns and enhance the adversarial robustness of image-processing-based defenses.

B. Generation of Spectral and Spatial Templates

The HSIs have 2-D spatial characteristics and 1-D spectral curves. To obtain smoother spectral–spatial perturbations, we use the superpixel segmentation method with SLIC to generate a 2-D oversegmented spatial template T_{spa} and apply the k-means clustering method to generate a 1-D oversegmented spectral template T_{spe}. We will promote the perturbations within a local spatial and spectral region to be similar.

The generation of spatial and spectral templates fully considers the land-cover characteristics, which can promote perturbation smoothing in the same land cover and avoid excessive smoothing of the perturbations in high-perturbation regions. Therefore, generating spatial and spectral templates by SLIC and the k-means methods can smooth perturbations while maintaining powerful attacks.

C. Attacks With Spectral–Spatial Consistency Constraints

In this section, the single-step FGSM is used as a baseline attack method, and the FGSM is optimized in

$$x^{\text{adv}} = \text{Clip}_{x,\varepsilon}\{x + \varepsilon \cdot \text{sign}(\nabla_x J(f(x), y))\} \tag{4}$$

where $\text{Clip}_{x,\varepsilon}$ means to crop the adversarial examples in $[x - \varepsilon, x + \varepsilon]$. Based on FGSM, we propose an enhanced FGSM method with a spectral–spatial consistency constraint (SS-FGSM), in which the object-level spatial and spectral

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perturbations are iteratively optimized under the constraint of the spatial and spectral templates.

1) Object-Level Spatial Attacks: The existing attack methods are mostly considered spatial attacks at the pixel level. We propose an object-level spatial attack method by utilizing the spatial template T_{spa}. The perturbations are initialized with random noise δ_0, and the object-level spatial perturbations are optimized according to

$$\begin{aligned}\delta_i^{m,n} &= \sum_{k\in m_N} \delta_i\left(T_{\text{spa}}^{m,k}\right)/m_N \\ \delta_{i+1} &= \text{Clip}_{\varepsilon}\left\{\delta_i + \varepsilon\cdot\text{sign}\left(\nabla_{\delta_i} J(f(x+\delta_i), y)\right)\right\} \\ x_{i+1}^{\text{adv}} &= \text{Clip}_{x,\varepsilon}\{x+\delta_{i+1}\}\end{aligned} \tag{5}$$

where $\sum_{k\in m_N}\delta_i(T_{\text{spa}}^{m,k})$ represents mapping the spatial template T_{spa} onto the perturbation map δ_i along the spatial dimensional and calculating the sum of spatial perturbations within the mth superpixel. m_N is the spatial pixel number within the mth superpixel. Therefore, $\delta_i^{m,n}$ represents that the nth perturbation within the mth superpixel is replaced by the average value of the spatial perturbations within this superpixel.

We first smooth the perturbations according to the spatial template and then update the perturbations according to the gradient of the classification model. The gradient calculation of most attack methods is performed on the input adversarial example x_i^{adv}, $x_i^{\text{adv}} = x+\delta_i$, while we directly calculate the gradient on perturbations δ_i. We update the perturbations rather than the example itself. Performing spatial smoothing on adversarial example x_i^{adv} can result in the loss of detailed information in the image itself, thereby affecting the classification accuracy of benign examples. Unreliable classification of benign examples can reduce the reliability of generated adversarial examples. Therefore, we obtain smoother perturbations instead of smoother adversarial examples.

2) Object-Level Spectral Attacks: To further reduce the influence of spectral attention differences on attack transferability, we generate object-level spectral perturbations by utilizing the spectral template T_{spe} after the object-level spatial attack, as shown in

$$\begin{aligned}\delta_i^{p,q} &= \sum_{z\in p_N} \delta_i\left(T_{\text{spe}}^{p,z}\right)/p_N \\ \delta_{i+1} &= \text{Clip}_{\varepsilon}\left\{\delta_i + \varepsilon\cdot\text{sign}\left(\nabla_{\delta_i} J(f(x+\delta_i), y)\right)\right\} \\ x_{i+1}^{\text{adv}} &= \text{Clip}_{x,\varepsilon}\{x+\delta_{i+1}\}\end{aligned} \tag{6}$$

where $\sum_{z\in p_N}\delta_i(T_{\text{spe}}^{p,z})$ represents mapping the spectral template T_{spe} onto the perturbation map δ_i along the spectral dimension and calculating the sum of spectral perturbations within the pth spectral subclass. p_N is the spectral pixel number in the pth spectral subclass. Therefore, $\delta_i^{p,q}$ denotes that the qth perturbation in the pth spectral subclass is replaced by the average of the spectral perturbations in this spectral subclass. Similar to (5), (6) smooths the perturbations along the spectral dimension based on the spectral template.

D. Incorporating With the Multistep Attack Methods

The proposed method can also be combined with other gradient-based multistep attack methods to further enhance their attack transferability and robustness.

Algorithm 1 Object-Level Adversarial Attack Method SS-FGSM.

Input: HSI example x, target model f, loss function J, perturbation budget δ, spatial superpixel size, spectral clustering size
Output: HSI adversarial example x^{adv}

$\delta_0 = \text{random}(C\times H\times W)$; $\delta_0^{adv} = x$
Calculate the spatial template T_{spa} and spectral template T_{spe}
for $i = 0 \to T-1$ **do**
  Update object-level spatial perturbations
    Smooth the spatial perturbation
      $\delta_i^{m,n} = \sum_{k\in m_N}\delta_i(T_{spa}^{m,k})/m_N$
    Update perturbation along the spatial-dimension
      $\delta_{i+1} = Clip_{\varepsilon}\{\delta_i + \varepsilon\cdot sign(\nabla_{\delta_i} J(f(x+\delta_i), y))\}$
  $i+1 \to i$
  Update object-level spectral perturbations
    Smooth the spectral perturbation
      $\delta_i^{p,q} = \sum_{z\in p_N}\delta_i(T_{spe}^{p,z})/p_N$
    Update perturbation along the spectral-dimension
      $\delta_{i+1} = Clip_{\varepsilon}\{\delta_i + \varepsilon\cdot sign(\nabla_{\delta_i} J(f(x+\delta_i), y))\}$
  Update HSI adversarial example $x_{i+1}^{adv} = Clip_{x,\varepsilon}\{x+\delta_{i+1}\}$
end
output $x^{adv} = x_T^{adv}$

Algorithm 2 Object-Level Adversarial Attack Method MI-SS-FGSM.

Input: HSI example x, target model f, loss function J, perturbation budget δ, step size α, decay factor μ, momentum term $g_{i,j}$, spatial superpixel size, spectral clustering size
Output: HSI adversarial example x^{adv}

$\delta_0 = \text{random}(C\times H\times W)$; $\delta_0^{adv} = x$; $g_0 = 0$
Calculate the spatial template T_{spa} and spectral template T_{spe}
for $i = 0 \to T-1$ **do**
  for $j = 0 \to t-1$ **do**
    Update object-level spatial perturbations
      Smooth the spatial perturbation
        $\delta_{i,j}^{m,n} = \sum_{k\in m_N}\delta_{i,j}(T_{spa}^{m,k})/m_N$
      Update perturbation along the spatial-dimension
        $g_{i,j+1} = \mu\cdot g_{i,j} + \frac{\nabla_{\delta_{i,j}}(f(x+\delta_{i,j}),y)}{\|\nabla_{\delta_{i,j}}(f(x+\delta_{i,j}),y)\|_1}$
        $\delta_{i,j+1} = Clip_{\varepsilon}\{\delta_{i,j} + \alpha\cdot sign(g_{i,j+1})\}$
  end
  for $j = 0 \to t-1$ **do**
    Update object-level spectral perturbations
      Smooth the spectral perturbation
        $\delta_{i,j}^{p,q} = \sum_{z\in p_N}\delta_{i,j}(T_{spe}^{p,z})/p_N$
      Update perturbation along the spectral-dimension
        $g_{i,j+1} = \mu\cdot g_{i,j} + \frac{\nabla_{\delta_{i,j}}(f(x+\delta_{i,j}),y)}{\|\nabla_{\delta_{i,j}}(f(x+\delta_{i,j}),y)\|_1}$
        $\delta_{i,j+1} = Clip_{\varepsilon}\{\delta_{i,j} + \alpha\cdot sign(g_{i,j+1})\}$
  end
  Update HSI adversarial example
    $x_{i+1,j+1}^{adv} = Clip_{x,\varepsilon}\{x+\delta_{i,j+1}\}$
end
output $x^{adv} = x_{T,t}^{adv}$

By incorporating the proposed high-dimensional spectral–spatial object-level perturbation generation framework with the MI-FGSM attack method, an extended MI-FGSM method with spectral–spatial consistency constraint (MI-SS-FGSM) is present to enhance the attack transferability and robustness of MI-FGSM. The multistep object-level spatial attack of MI-SS-FGSM is defined in

$$\delta_{i,j}^{m,n} = \sum_{k\in m_N}\delta_{i,j}\left(T_{\text{spa}}^{m,k}\right)/m_N$$

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$$
\begin{aligned}
g_{i,j+1} &= \mu \cdot g_{i,j} + \frac{\nabla_{\delta_{i,j}}\left(f\left(x+\delta_{i,j}\right), y\right)}{\left\|\nabla_{\delta_{i,j}}\left(f\left(x+\delta_{i,j}\right), y\right)\right\|_1} \\
\delta_{i,j+1} &= \mathrm{Clip}_{\varepsilon}\left\{\delta_{i,j} + \alpha \cdot \mathrm{sign}\left(g_{i,j+1}\right)\right\} \\
x^{\mathrm{adv}}_{i+1,j+1} &= \mathrm{Clip}_{x,\varepsilon}\left\{x + \delta_{i,j+1}\right\}
\end{aligned} \tag{7}
$$

where i represents the outer loop to promote iterative optimization of the spatial and spectral perturbations and j represents the inner loop to implement a multistep object-level spatial attack. $g_{i,j}$ denotes the momentum term, $g_0 = 0$, μ is the decay factor, and α is the step size. The perturbations δ_0 are initialized with random noise. The multistep object-level spectral perturbation has a similar representation.

The VMI-FGSM is used as the baseline multistep attack method and an extended VMI-FGSM method with a spectral–spatial consistency constraint (VMI-SS-FGSM) is proposed, and the multistep object-level spatial attack of VMI-SS-FGSM is defined in

$$
\begin{aligned}
\delta^{m,n}_{i,j} &= \sum_{k \in m_N} \delta_{i,j}\left(T^{m,k}_{\mathrm{spa}}\right)/m_N \\
g_{i,j+1} &= \mu \cdot g_{i,j} + \frac{\nabla_{\delta_{i,j}}\left(f\left(x+\delta_{i,j}\right), y\right) + v_i}{\left\|\nabla_{\delta_{i,j}}\left(f\left(x+\delta_{i,j}\right), y\right) + v_i\right\|_1} \\
v_{i,j+1} &= V\left(\delta_{i,j}\right) \\
\delta_{i+1,j+1} &= \mathrm{Clip}_{\varepsilon}\left\{\delta_{i,j} + \alpha \cdot \mathrm{sign}\left(g_{i,j+1}\right)\right\} \\
x^{\mathrm{adv}}_{i+1,j+1} &= \mathrm{Clip}_{x,\varepsilon}\left\{x + \delta_{i+1,j+1}\right\}
\end{aligned} \tag{8}
$$

where $v_0 = 0$, $V(\delta_{i,j}) = (1/N)\sum_{v=1}^{N} \nabla_{\delta^v_{i,j}} J(f(x+\delta^v_{i,j}), y) - \nabla_{\delta_{i,j}} J(f(x+\delta_{i,j}), y)$ and $\delta^j_i = \delta_i + r_i, r_i \sim U[-(\beta \cdot \varepsilon)^d, (\beta \cdot \varepsilon)^d]$, where β is the upper bound of neighborhood.

IV. Experiments

In this section, we conduct experiments on two real HSI datasets to verify the effectiveness of our method.

A. Experimental Settings

1) Datasets: The HoustonU 2018 and PaviaU datasets are used in the experiments, as shown in Fig. 3.

The HoustonU 2018 dataset was acquired by the HSI CASI 1500 over the region of the University of Houston on February 16, 2017. The spatial resolution is 1 m and the spatial size of HoustonU 2018 is 601 × 2384 with 48 spectral bands. The dataset contains 20 available classes of interest.

The PaviaU dataset was acquired by the ROSIS-03 satellite over the region of the University of Pavia. The spatial resolution is 1.3 m and the spatial size of Pavia University is 610 × 304 with 115 spectral bands. Twelve noisy spectral bands were removed before the experiments, and nine available classes were used for classification.

2) Training Examples: For each pixel, a neighborhood window with spatial size 32 × 32 is applied to construct the HSI examples. The detailed number of training and testing examples is shown in Table I.

3) Networks: Five trained classification networks are used to verify the attack transferability, that is, VGG-11, Resnet-18, VGG-19, Inception v3 (Inc-v3) [43], and Inception Resnet V2 (IncRes-v2) [44]. One of the five models is used as a white-box model to generate adversarial examples, and the other models are used to test attack transferability. In addition, we apply two well-known image-processing-based defense methods to verify the attack robustness, including spatial resizing and spectral smoothing.

4) Attacking Methods: We use FGSM [12], MI-FGSM [13], and VMI-FGSM [17] as baseline attack methods and combine our method with the baseline attack methods, named SS-FGSM, MI-SS-FGSM, and VMI-SS-FGSM, to evaluate the attack transferability and robustness. Additionally, the most related work in [9] is also combined with the baseline attack methods for the comparisons, which are named SA-FGSM, MI-SA-FGSM, and VMI-SA-FGSM.

5) Parameters: The higher attack success rate typically demands a larger perturbation budget, but such perturbations are easily detected by the human visual system. Therefore, enhancing the attack power of adversarial samples with a small perturbation budget is a challenging problem. In this article, smaller perturbation budgets of 0.01 and 0.03 are chosen to

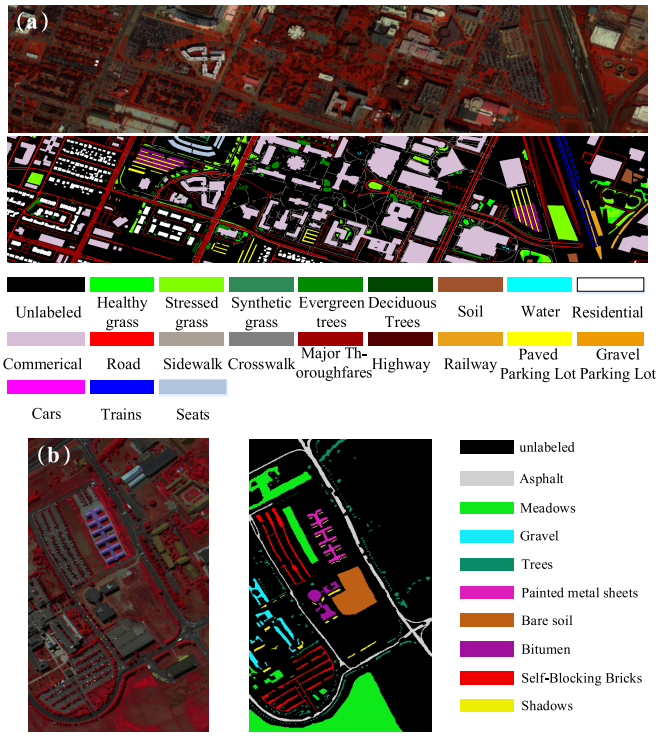


Fig. 3. False color images and ground-truth maps of the (a) HoustonU 2018 and (b) PaviaU datasets.

TABLE I

Training and Testing Sets of the HoustonU 2018 and PaviaU Datasets

HoustonU 2018						PaviaU		
class	Train	Test	class	Train	Test	class	Train	Test
1	5000	4399	11	10000	24029	1	900	5731
2	10000	22502	12	1000	518	2	900	17749
3	500	184	13	10000	36348	3	900	1199
4	10000	3595	14	5000	4865	4	900	2164
5	3000	2021	15	5000	1937	5	900	445
6	3000	1516	16	10000	1500	6	900	4129
7	200	66	17	100	46	7	900	430
8	10000	29772	18	3000	3547	8	900	2782
9	10000	213752	19	3000	2369	9	900	47
10	10000	35866	20	3000	3824			

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TABLE II

CLASSIFICATION ACCURACY (%) OF ADVERSARIAL ATTACKS AGAINST FIVE TRAINED NETWORKS ON THE HOUSTONU 2018 DATASET. THE ADVERSARIAL EXAMPLES ARE CRAFTED ON VGG-11. THE MAXIMUM PERTURBATION BUDGET IS SET TO 0.01 AND 0.03, RESPECTIVELY

Attack method	Perturbation budget	VGG-11		ResNet-18		VGG-19		Inc-v3		IncRes-v2	
		0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
NONE	OA	99.01	99.01	99.20	99.20	99.24	99.24	99.41	99.41	99.17	99.17
	AA	99.11	99.11	99.36	99.36	99.24	99.24	99.53	99.53	99.33	99.33
	Kappa	99.00	99.00	99.15	99.15	99.19	99.19	99.38	99.38	99.12	99.12
FGSM[12]	OA	24.75	2.83	89.35	22.35	76.14	15.63	84.51	15.92	74.56	16.92
	AA	27.22	2.52	90.55	23.52	79.48	16.37	85.94	17.01	76.64	19.01
	Kappa	20.01	-3.54	88.53	17.29	74.66	6.94	83.53	10.46	72.91	11.55
SA-FGSM[9]	OA	29.16	2.51	72.40	19.71	54.20	13.68	71.03	16.42	**52.21**	15.72
	AA	29.86	1.09	72.96	20.46	55.69	13.69	70.95	17.13	**51.85**	16.84
	Kappa	25.19	0.16	70.46	14.86	50.75	13.97	69.41	14.96	**50.95**	9.67
SS-FGSM	OA	**10.10**	**1.32**	**70.37**	**18.82**	**51.58**	**12.30**	**70.90**	**13.97**	53.41	**13.73**
	AA	**12.80**	**1.00**	**70.22**	**20.29**	**53.47**	**11.56**	**70.71**	**15.19**	54.11	**15.13**
	Kappa	**4.48**	**-5.12**	**72.11**	**13.57**	**49.75**	**6.61**	**68.93**	**8.06**	53.69	**8.17**
MI-FGSM[13]	OA	3.43	0.84	82.72	18.12	68.37	10.76	80.58	11.88	65.30	13.06
	AA	**3.59**	0.56	84.91	20.62	72.81	10.26	83.53	13.54	69.33	15.35
	Kappa	**-3.75**	-6.53	81.50	13.07	66.43	3.77	79.55	6.15	63.04	7.31
MI-SA-FGSM[9]	OA	13.08	2.25	74.31	15.63	55.59	10.53	75.13	11.47	**54.25**	11.61
	AA	12.49	1.50	73.83	16.11	**55.11**	8.96	71.66	12.44	55.90	13.79
	Kappa	11.44	-4.55	73.78	9.55	**52.68**	4.41	71.57	5.37	52.81	6.03
MI-SS-FGSM	OA	**2.74**	**0.02**	**74.18**	**13.68**	**51.67**	**7.68**	**74.97**	**10.45**	55.09	**9.73**
	AA	4.51	**0.01**	**72.19**	**15.43**	56.82	**7.94**	**70.04**	**12.23**	**55.39**	**12.32**
	Kappa	-3.37	**-6.53**	**72.52**	**7.69**	52.75	**1.41**	**70.28**	**4.55**	**52.40**	**3.97**
VMI-FGSM[17]	OA	5.82	**0.11**	83.04	**18.38**	69.26	12.41	64.93	18.17	67.39	15.62
	AA	8.49	**0.25**	81.83	**19.11**	55.11	8.96	64.66	16.44	67.90	16.79
	Kappa	1.44	**-4.55**	81.78	**12.55**	54.68	4.41	64.57	13.37	65.81	14.03
VMI-SA-FGSM[9]	OA	4.69	1.25	**69.43**	33.74	52.89	16.52	50.38	21.39	55.34	20.87
	AA	8.64	1.06	**68.46**	33.65	52.56	13.01	51.49	22.42	52.50	14.68
	Kappa	0.63	0.04	**65.94**	29.56	50.16	12.78	47.66	18.46	50.31	**11.95**
VMI-SS-FGSM	OA	**3.20**	0.41	69.81	29.30	**48.13**	**11.08**	**45.48**	**16.14**	**52.91**	**14.13**
	AA	**5.43**	0.91	69.45	31.95	**47.62**	**8.12**	**45.23**	**14.69**	**52.19**	**10.90**
	Kappa	**0.05**	-0.92	66.72	22.47	**43.94**	**3.63**	**43.82**	**12.06**	**50.13**	13.17

assess the effectiveness of the proposed method. Additionally, the spatial superpixel size is set to four spatial pixels, and the spectral clustering size is set to 20 spectral pixels. The spatial clustering size in ground sample distances (GSDs) is 4 m for the HoustonU 2018 dataset and 5.2 m for the PaviaU dataset. The outer iteration number $T = 20$, the inter iteration number $t = 5$, $\mu = 1$, $\alpha = \delta(\mathrm{NC})^{1/2}/t$, and $\mathrm{NC} = C \times H \times W$ [9].

B. Attack Transferability

In this section, we evaluate the performance of our method on five pretrained classification models. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to evaluate the attack performance. The experimental results on the HoustonU 2018 dataset are presented in Tables II and III, while the results on the PaviaU dataset are shown in Tables IV and V. We report the classification accuracy of the models when subjected to adversarial attacks with maximum perturbation budgets of 0.01 and 0.03.

1) HoustonU 2018 Dataset: In Tables II and III, the HSI adversarial examples are generated using the white-box models VGG-11 and ResNet-18, respectively. It can be observed that our methods effectively preserve or even enhance the attack success rate on white-box models. The advantages of white-box attacks are particularly prominent in the case of FGSM. With a 0.01 perturbation budget, the classification accuracy is reduced by nearly 15% on both VGG-11 and ResNet-18 models. Furthermore, when our methods are combined with the baseline attacks and transferred to attack other four trained black-box models, a significant decrease in classification accuracy is observed. This decrease is especially pronounced when combined with FGSM, resulting in improved attack transferability. As shown in Table II, the OA value of SS-FGSM is reduced by nearly 20% compared to the baseline FGSM attack. Similarly, for the multistep attack methods MI-FGSM and VMI-FGSM, our MI-SS-FGSM and VMI-SS-FGSM achieve a reduction of nearly 10% in OA metrics. While the SA-FGSM, MI-SA-FGSM, and VMI-SA-FGSM outperform the baseline attacks in terms of attack transferability, their overall attack success rate is less satisfactory. In summary, our methods effectively maintain or improve the attack performance on white-box models, while enhancing the attack transferability on black-box models.

2) PaviaU Dataset: In Tables IV and V, we present the results obtained when generating adversarial examples on VGG-11 and ResNet-18 models with maximum perturbation budgets of 0.01 and 0.03, respectively. The remaining classification models are considered black-box models. Unlike the HoustonU 2018 dataset, attacking the black-box models on this dataset is more challenging due to the increased number of spectral bands. Nevertheless, our methods still demonstrate effective improvement in the attack transferability of the baseline attacks in most cases. Despite the increased difficulty, we can enhance the attack transferability on the black-box models.

Additionally, we analyze the computational complexity of the compared attack methods, as shown in Tables VI and VII. The testing time is the time to generate adversarial samples for one test batch (batch size is 1000). From the perspective of model structure, the attack time is closely related to the

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TABLE III

CLASSIFICATION ACCURACY (%) OF ADVERSARIAL ATTACKS AGAINST FIVE TRAINED NETWORKS ON THE HOUSTONU 2018 DATASET. THE ADVERSARIAL EXAMPLES ARE CRAFTED ON RESNET-18. THE MAXIMUM PERTURBATION BUDGET IS SET TO 0.01 AND 0.03, RESPECTIVELY

Attack method	Perturbation budget	ResNet-18		VGG-11		VGG-19		Inc-v3		IncRes-v2	
		0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
NONE	OA	99.20	99.20	99.01	99.01	99.24	99.24	99.41	99.41	99.17	99.17
	AA	99.36	99.36	99.11	99.11	99.24	99.24	99.53	99.53	99.33	99.33
	Kappa	99.15	99.15	99.00	99.00	99.19	99.19	99.38	99.38	99.12	99.12
FGSM[12]	OA	24.14	14.96	95.82	56.07	92.19	41.52	87.83	29.79	90.76	44.30
	AA	24.80	14.41	95.77	51.72	92.19	44.06	90.05	28.75	90.95	48.00
	Kappa	19.00	8.72	95.54	52.82	89.05	37.12	87.05	24.42	90.09	40.06
SA-FGSM[9]	OA	50.47	7.41	90.92	**31.18**	79.18	25.61	84.64	25.87	**75.12**	**27.32**
	AA	52.36	9.81	**82.89**	**32.07**	80.24	25.80	83.33	26.16	**75.99**	**21.57**
	Kappa	47.99	5.50	**87.46**	**30.76**	**76.12**	23.46	81.44	23.79	**77.53**	**25.76**
SS-FGSM	OA	**11.43**	**3.76**	**89.94**	33.68	**78.75**	**25.28**	**84.25**	**22.90**	84.94	29.86
	AA	**14.16**	**3.90**	89.84	36.71	**80.11**	**24.92**	**82.37**	**24.05**	84.29	22.44
	Kappa	**6.26**	**-2.29**	93.95	31.14	82.43	**21.35**	**81.29**	**18.39**	86.49	26.45
MI-FGSM[13]	OA	10.32	9.99	94.93	53.51	88.84	37.28	86.90	29.52	89.45	43.51
	AA	10.22	8.17	95.65	50.30	89.95	35.07	87.96	28.78	89.77	41.48
	Kappa	3.79	3.34	94.66	49.82	88.16	32.49	86.09	24.26	88.79	39.07
MI-SA-FGSM[9]	OA	11.94	8.61	94.24	**37.39**	79.55	25.79	81.67	20.83	80.39	**33.14**
	AA	11.35	9.87	93.77	39.26	**79.15**	29.63	80.31	22.53	80.72	35.14
	Kappa	10.42	2.01	92.57	**33.41**	77.71	20.98	81.81	16.31	**77.04**	**28.53**
MI-SS-FGSM	OA	**9.92**	**5.90**	**94.05**	39.43	**78.38**	**24.84**	**80.41**	**19.79**	**80.31**	34.12
	AA	**10.19**	**4.73**	**89.18**	**38.28**	80.06	**24.53**	**80.09**	**19.40**	**78.00**	**33.59**
	Kappa	**3.68**	**-0.90**	**91.75**	35.15	**77.04**	**20.86**	**79.37**	**16.17**	81.63	29.61
VMI-FGSM[17]	OA	**2.80**	**0.12**	93.95	38.58	83.22	**26.32**	74.38	22.43	85.20	32.31
	AA	2.35	0.87	92.77	39.26	83.15	29.63	70.31	22.53	86.72	31.14
	Kappa	0.05	**-0.59**	90.57	33.41	78.71	**19.84**	71.81	**16.31**	83.04	29.53
VMI-SA-FGSM[9]	OA	7.28	0.65	79.60	18.75	72.42	30.48	**60.19**	28.12	74.63	35.93
	AA	1.90	0.56	80.62	18.04	72.92	30.86	**59.04**	27.26	73.01	35.16
	Kappa	0.10	-0.10	75.14	13.96	69.41	26.37	**54.98**	26.17	72.65	30.65
VMI-SS-FGSM	OA	5.22	0.40	**76.50**	**16.77**	**68.27**	29.36	60.25	**21.00**	**72.89**	**32.21**
	AA	**0.99**	**0.01**	**76.05**	**15.63**	**67.41**	**28.39**	62.46	**22.15**	**72.46**	**30.78**
	Kappa	**-0.13**	-0.55	**74.66**	**12.75**	**63.89**	25.75	58.93	19.84	**70.79**	**28.65**

TABLE IV

CLASSIFICATION ACCURACY (%) OF ADVERSARIAL ATTACKS AGAINST FIVE TRAINED NETWORKS ON THE PAVIAU DATASET. THE ADVERSARIAL EXAMPLES ARE CRAFTED ON VGG-11. THE MAXIMUM PERTURBATION BUDGET IS SET TO 0.01 AND 0.03, RESPECTIVELY

Attack method	Perturbation budget	VGG-11		ResNet-18		VGG-19		Inc-v3		IncRes-v2	
		0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
NONE	OA	99.93	99.93	99.99	99.99	99.51	99.51	99.94	99.94	99.52	99.52
	AA	99.94	99.94	99.98	99.98	99.58	99.58	99.96	99.96	99.83	99.83
	Kappa	99.92	99.92	99.97	99.97	99.43	99.43	99.93	99.93	99.77	99.77
FGSM[12]	OA	61.67	14.42	72.36	43.44	70.60	49.52	99.58	75.20	85.93	56.17
	AA	63.86	22.98	76.90	49.49	72.91	53.29	99.68	74.75	88.27	59.97
	Kappa	55.62	0.80	68.06	34.38	65.94	41.47	99.52	71.31	83.74	49.15
SA-FGSM[9]	OA	65.00	16.99	69.50	27.58	67.48	42.44	98.76	60.58	**79.04**	34.54
	AA	68.01	24.61	**64.56**	37.44	70.83	49.93	99.16	**66.48**	**82.27**	44.59
	Kappa	60.11	3.83	**54.58**	14.36	63.15	33.51	98.83	**55.26**	**76.47**	23.86
SS-FGSM	OA	**49.31**	**7.94**	**62.57**	**25.38**	**67.45**	**37.50**	**98.12**	**60.06**	81.99	**33.41**
	AA	**55.48**	**16.68**	66.16	**35.55**	**70.45**	**46.91**	**99.10**	69.49	83.76	**43.09**
	Kappa	**49.99**	**-4.23**	56.62	**14.13**	**62.34**	**27.61**	**98.75**	60.06	78.17	**22.76**
MI-FGSM[13]	OA	**46.94**	**4.49**	68.35	31.93	69.62	35.00	99.12	65.21	82.27	43.54
	AA	**51.70**	15.04	71.82	40.93	72.30	42.63	99.33	69.51	84.26	59.66
	Kappa	**38.61**	**-10.52**	63.32	20.98	64.82	24.64	99.00	59.66	79.49	34.59
MI-SA-FGSM[9]	OA	66.25	18.75	69.09	28.20	69.80	41.69	98.86	**58.62**	**80.48**	37.67
	AA	68.42	26.75	**68.67**	37.98	60.87	49.28	99.20	**52.85**	**83.85**	45.44
	Kappa	60.87	6.16	**57.75**	16.12	63.32	32.85	**98.80**	**53.20**	**77.87**	26.64
MI-SS-FGSM	OA	49.21	4.58	**66.01**	**22.31**	**68.23**	**32.20**	**98.16**	61.88	80.72	**36.45**
	AA	54.00	**14.85**	69.04	**34.66**	**60.82**	**40.60**	**99.14**	65.90	83.91	**44.16**
	Kappa	41.63	-10.10	60.66	**12.98**	**63.27**	**21.53**	99.02	56.39	78.96	**25.79**
VMI-FGSM[17]	OA	56.78	**8.51**	70.31	40.08	68.71	43.10	93.63	**67.41**	83.70	46.72
	AA	58.33	16.60	73.65	44.53	70.90	46.90	99.51	71.76	85.18	52.52
	Kappa	47.79	**-7.13**	64.80	36.91	63.50	**31.83**	99.27	65.36	80.17	**37.38**
VMI-SA-FGSM[9]	OA	52.87	17.91	66.69	40.57	67.94	46.72	86.30	69.34	73.63	44.76
	AA	65.14	19.57	64.34	34.01	70.18	46.33	98.99	**67.92**	80.35	37.43
	Kappa	56.46	-2.36	**53.98**	32.13	61.94	38.39	98.55	65.53	74.85	44.74
VMI-SS-FGSM	OA	**46.50**	13.61	**63.07**	**33.62**	**66.61**	**41.84**	**81.74**	69.20	**68.35**	**43.27**
	AA	**46.09**	**13.18**	**57.25**	**31.06**	**62.57**	**38.58**	**77.36**	70.23	**58.14**	**36.09**
	Kappa	**38.56**	8.48	60.24	**25.76**	**60.35**	34.51	**79.81**	**63.43**	**67.09**	44.83

computational and parameter quantities of the target model. ResNet-18 has more computational and parameter quantities than VGG-11, thus it takes more time to attack. From the perspective of attack methods, compared to baseline attack

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TABLE V

CLASSIFICATION ACCURACY (%) OF ADVERSARIAL ATTACKS AGAINST FIVE TRAINED NETWORKS ON THE PAVIAU DATASET. THE ADVERSARIAL EXAMPLES ARE CRAFTED ON RESNET-18. THE MAXIMUM PERTURBATION BUDGET IS SET TO 0.01 AND 0.03, RESPECTIVELY

Attack method	Perturbation budget	ResNet-18		VGG-11		VGG-19		Inc-v3		IncRes-v2	
		0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
NONE	OA	99.99	99.99	99.93	99.93	99.51	99.51	99.94	99.94	99.52	99.52
	AA	99.98	99.98	99.94	99.94	99.58	99.58	99.96	99.96	99.83	99.83
	Kappa	99.97	99.97	99.92	99.92	99.43	99.43	99.93	99.93	99.77	99.77
FGSM[12]	OA	50.14	16.77	82.07	56.34	82.10	56.34	99.19	71.43	88.82	52.43
	AA	55.49	25.46	85.25	60.14	80.51	61.52	99.38	71.70	88.72	57.15
	Kappa	42.23	3.66	79.30	56.34	74.70	51.36	99.07	66.94	87.07	44.90
SA-FGSM[9]	OA	42.85	10.63	72.57	40.64	69.49	47.01	98.88	**60.06**	**78.07**	36.43
	AA	50.21	20.49	**76.49**	48.92	73.04	53.56	98.33	**61.96**	**79.25**	44.17
	Kappa	34.31	-4.41	**68.33**	31.68	64.81	38.40	97.90	**52.46**	**74.51**	**26.10**
SS-FGSM	OA	**29.20**	**3.81**	**72.46**	**40.55**	**69.45**	**45.04**	**98.19**	61.32	80.24	**33.90**
	AA	**39.44**	**12.50**	77.48	**48.11**	**72.82**	**53.15**	**98.31**	63.69	81.01	**35.99**
	Kappa	**29.79**	**-10.42**	69.67	**31.18**	**64.63**	**37.93**	**97.53**	54.48	77.12	29.83
MI-FGSM[13]	OA	28.53	2.09	78.43	50.65	72.25	52.14	98.63	63.42	84.37	46.22
	AA	37.82	3.71	80.99	53.66	75.35	55.34	98.86	63.74	84.16	52.81
	Kappa	17.35	-13.52	74.24	41.30	67.78	42.04	98.28	54.71	81.29	39.11
MI-SA-FGSM[9]	OA	45.95	11.43	72.62	38.77	69.90	46.23	98.32	56.40	77.67	35.41
	AA	52.96	22.52	78.30	**46.01**	73.40	52.81	98.70	59.38	**78.95**	**43.55**
	Kappa	38.39	-3.26	70.33	**29.55**	65.55	38.79	98.05	48.60	**74.55**	**25.07**
MI-SS-FGSM	OA	**25.86**	**1.20**	**71.52**	**37.01**	**69.65**	**45.80**	**97.91**	**55.89**	**76.36**	**34.30**
	AA	**36.26**	**2.28**	**77.24**	47.67	**73.04**	**51.99**	**98.37**	**59.30**	82.07	47.75
	Kappa	**14.58**	**-14.63**	**69.46**	32.07	**64.83**	**37.70**	**97.55**	**48.49**	78.36	31.47
VMI-FGSM[17]	OA	33.05	4.61	75.51	49.74	71.70	51.69	91.90	64.98	82.58	46.37
	AA	40.56	7.78	80.63	52.69	74.28	55.02	98.77	62.47	83.85	50.15
	Kappa	20.91	-11.23	73.15	39.71	66.34	41.82	98.15	53.09	80.49	35.26
VMI-SA-FGSM[9]	OA	31.64	5.35	69.73	47.66	68.42	51.55	90.64	62.32	74.60	**43.15**
	AA	44.56	14.97	73.64	41.50	71.75	52.18	97.46	59.42	76.06	42.69
	Kappa	26.04	-8.12	65.95	33.68	63.87	**40.72**	96.22	52.62	70.67	**23.87**
VMI-SS-FGSM	OA	**22.97**	**3.27**	**68.30**	**47.37**	**66.47**	**51.23**	**85.54**	**57.82**	**68.35**	43.43
	AA	**20.21**	**0.02**	**50.26**	**37.04**	**60.30**	**39.46**	**80.61**	**48.94**	**56.70**	**39.57**
	Kappa	**15.80**	**-16.01**	**55.10**	**32.10**	**60.23**	41.49	**75.22**	**44.95**	**66.60**	29.13

TABLE VI

COMPUTATIONAL COMPLEXITY ANALYSIS ON THE HOUSTONU 2018 DATASET [RUNNING TIME (S)]

Attacks/ Target model*	FGSM	SS-FGSM	MI-FGSM	MI-SS-FGSM	VMI-FGSM	VMI-SS-FGSM
VGG11	0.11	15.28	0.75	78.99	12.43	520.11
ResNet18	0.23	19.33	0.86	102.74	30.27	821.58

* Flops VGG11: 171.59M ResNet18: 557.35M
Params VGG11: 8.83M ResNet18: 10.69M
Testbatchsize 1000 / GPU: 3090

TABLE VII

COMPUTATIONAL COMPLEXITY ANALYSIS ON THE PAVIAU DATASET [RUNNING TIME (S)]

Attacks/ Target model*	FGSM	SS-FGSM	MI-FGSM	MI-SS-FGSM	VMI-FGSM	VMI-SS-FGSM
VGG11	0.13	15.56	0.89	79.98	17.50	881.54
ResNet18	0.39	25.69	2.11	129.53	45.01	1831.21

* Flops VGG11: 202.53M ResNet18: 588.29M
Params VGG11: 8.86M ResNet18: 10.72M
Testbatchsize 1000 / GPU: 3090

methods, the proposed method increases the number of iterations and needs to smooth the spatial and spectral perturbations in each iteration, and therefore, it takes more time than the baseline methods.

C. Attack Advanced Defense Models

1) Spatial Resizing: Spatial resizing can effectively reduce local perturbations by linear interpolation. Assume that the size of the HSI adversarial example is $C \times H \times W$. We first perform a spatial downsampling operator with a resize factor of κ followed by a spatial upsampling operator to achieve spatial resizing-based defense. Fig. 4 illustrates the classification accuracy of HSI adversarial examples with different resize factors ranging from 1.0 to 3.0. The results demonstrate that larger-size spatial resizing leads to a more significant improvement in classification accuracy as it removes more perturbations. However, compared to the baseline attack methods, our proposed method exhibits more advantages in terms of large-size spatial resizing, and the change curve shows a gentler trend. In comparison to the SA-FGSM, MI-SA-FGSM, and VMI-SA-FGSM, our method achieves similar defense effects by promoting perturbation smoothing iteratively, rather than forcing perturbation smoothing. As a result, our method demonstrates greater robustness to spatial resizing.

2) Spectral Filtering: The HSI has redundant spectral bands; hence, local spectral filtering is also a challenging

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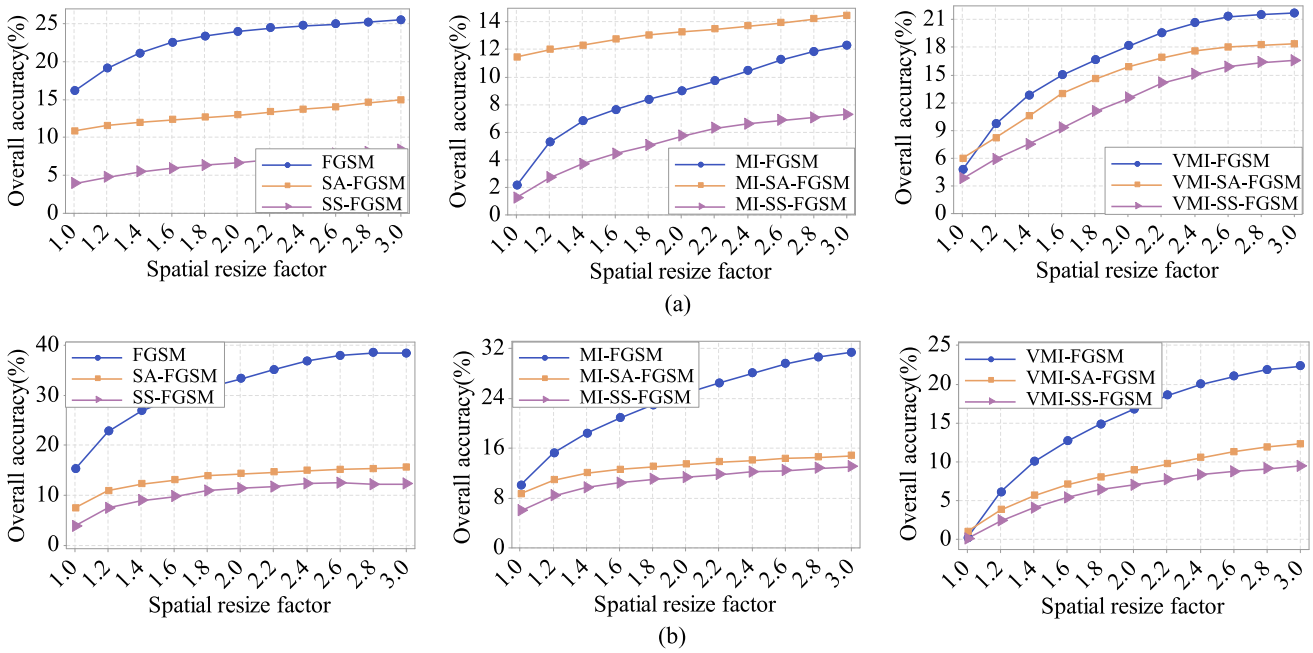


Fig. 4. Classification accuracy of HSI adversarial examples with image resizing. The HSI adversarial examples are generated on ResNet-18, and the maximum perturbation budget is 0.03. The range of the resize factor is 1.0–3.0. (a) PaviaU. (b) HoustonU 2018.

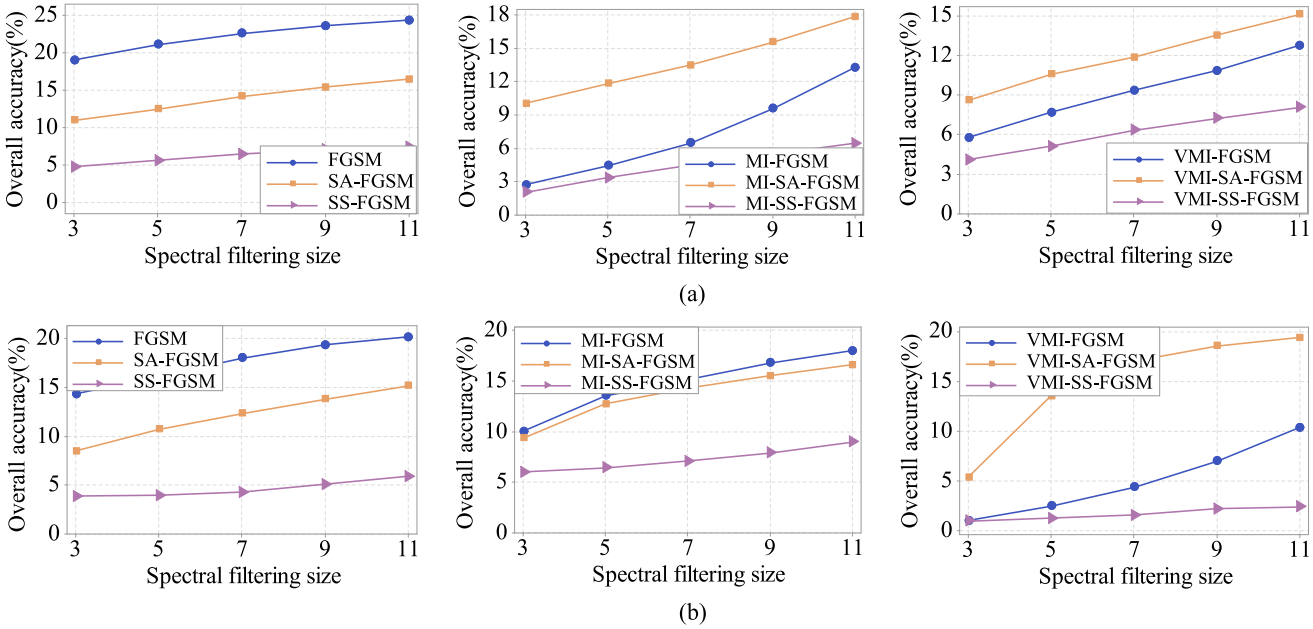


Fig. 5. Classification accuracy of HSI adversarial examples with spectral filtering. The HSI adversarial examples are generated on ResNet-18, and the maximum perturbation budget is 0.03. The filter size ranges from 3 to 7. (a) PaviaU. (b) HoustonU 2018.

defense method. We perform mean filtering along the spectral dimension to verify the robustness of the attack methods. The change curve is shown in Fig. 5, and the filter size ranges from 3 to 11. Our proposed method exhibits significant advantages, particularly in large-scale filtering scenarios. For example, when the filter size is 3, the VMI-FGSM and VMI-SS-FGSM methods show similar classification accuracy. However, when the filter size increases to 11, the classification accuracy decreases to 10.46% for VMI-FGSM and only 2.36% for VMI-SS-FGSM. This difference arises from the full consideration of spectral local smoothing characteristics in our proposed method. By incorporating spectral local smoothing through our approach, we achieve improved robustness against spectral filtering defenses. This highlights the effectiveness of our method in preserving adversarial perturbations even under large-scale spectral filtering operations.

D. Ablation Study on Hyperparameters

1) Ablation Study on Spatial and Spectral Consistency Constraints: The ablation study on spatial and spectral consistency

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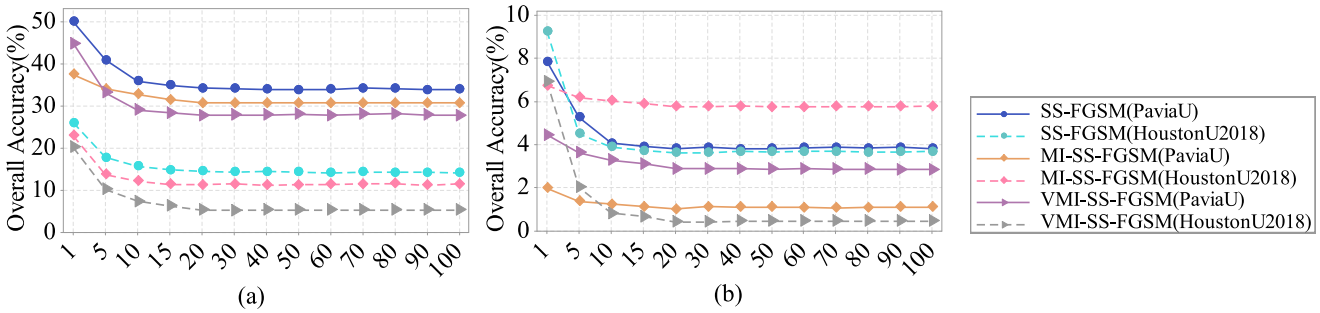


Fig. 6. Classification accuracy with the number of different iterations. (a) Iteration number (perturbation budge 0.01). (b) Iteration number (perturbation budge 0.03).

TABLE VIII

ABLATION STUDY ON SPATIAL AND SPECTRAL CONSISTENCY CONSTRAINTS. THE ADVERSARIAL EXAMPLES ARE CRAFTED ON RESNET-18. THE PERTURBATION BUDGET IS SET TO 0.03

Constrain condition			HoustonU 2018		
Spatial	Spectral	iteration number	SS-FGSM	MI-SS-FGSM	VMI-SS-FGSM
✓		T=1	14.51	10.76	16.88
	✓	T=1	12.79	7.43	19.04
✓	✓	T=1	9.24	6.62	6.91
✓	✓	T=20	3.76	5.90	0.40
Constrain condition			PaviaU		
Spatial	Spectral	iteration number	SS-FGSM	MI-SS-FGSM	VMI-SS-FGSM
✓		T=1	16.86	4.62	13.02
	✓	T=1	22.06	2.85	7.35
✓	✓	T=1	7.84	1.98	4.25
✓	✓	T=20	3.81	1.20	3.27

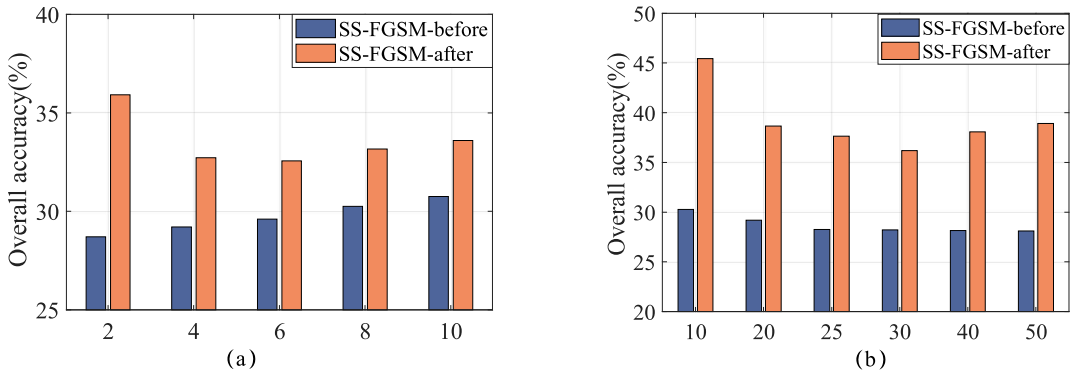


Fig. 7. (a) OA with different superpixel sizes on the PaviaU dataset. The adversarial examples are crafted on ResNet-18, the spectral cluster size is 20, and the spatial resizing factor is 2. (b) Classification accuracy with different spectral clustering sizes. The adversarial examples are crafted on ResNet-18, the superpixel size is 4, and the filtering size is 7×7.

constraints provides insights into their influences on attack ability. The analysis results are summarized in Table VIII and Fig. 6. Table VIII presents four different conditions considered for constraining perturbations: spatial constraint, spectral constraint, both spatial and spectral constraints, and iteration number. Two cases are examined: when the iteration number is 1 and when the iteration number is 20. It is observed that constraining both spatial and spectral perturbations yields significantly better attack effects compared to constraining only one of them. Furthermore, increasing the iteration number further improves the attack success rate.

Additionally, the classification accuracies with different iteration numbers are tested and illustrated in Fig. 6. The curves show that when the iteration number exceeds 20, the accuracy improvements become marginal. Therefore, setting the number of iterations to 20 is a more appropriate choice, considering both accuracy and time costs.

Overall, the ablation study confirms the effectiveness of spatial and spectral consistency constraints in enhancing the attack ability. Constraining both spatial and spectral perturbations and performing a sufficient number of iterations significantly improve the attack success rate while balancing the tradeoff between accuracy and computational efficiency.

2) Spatial Superpixel Size: In this section, the influence of superpixel size on attack performance is analyzed, as shown in Fig. 7(a). The number of spatial pixels within each superpixel ranges from 2 to 10. We test the adversarial robustness before and after spatial resizing. Before spatial resizing, large-scale superpixels lead to a decrease in attack ability, while after spatial resizing, the classification accuracy first decreases and then increases. Based on these observations, it is

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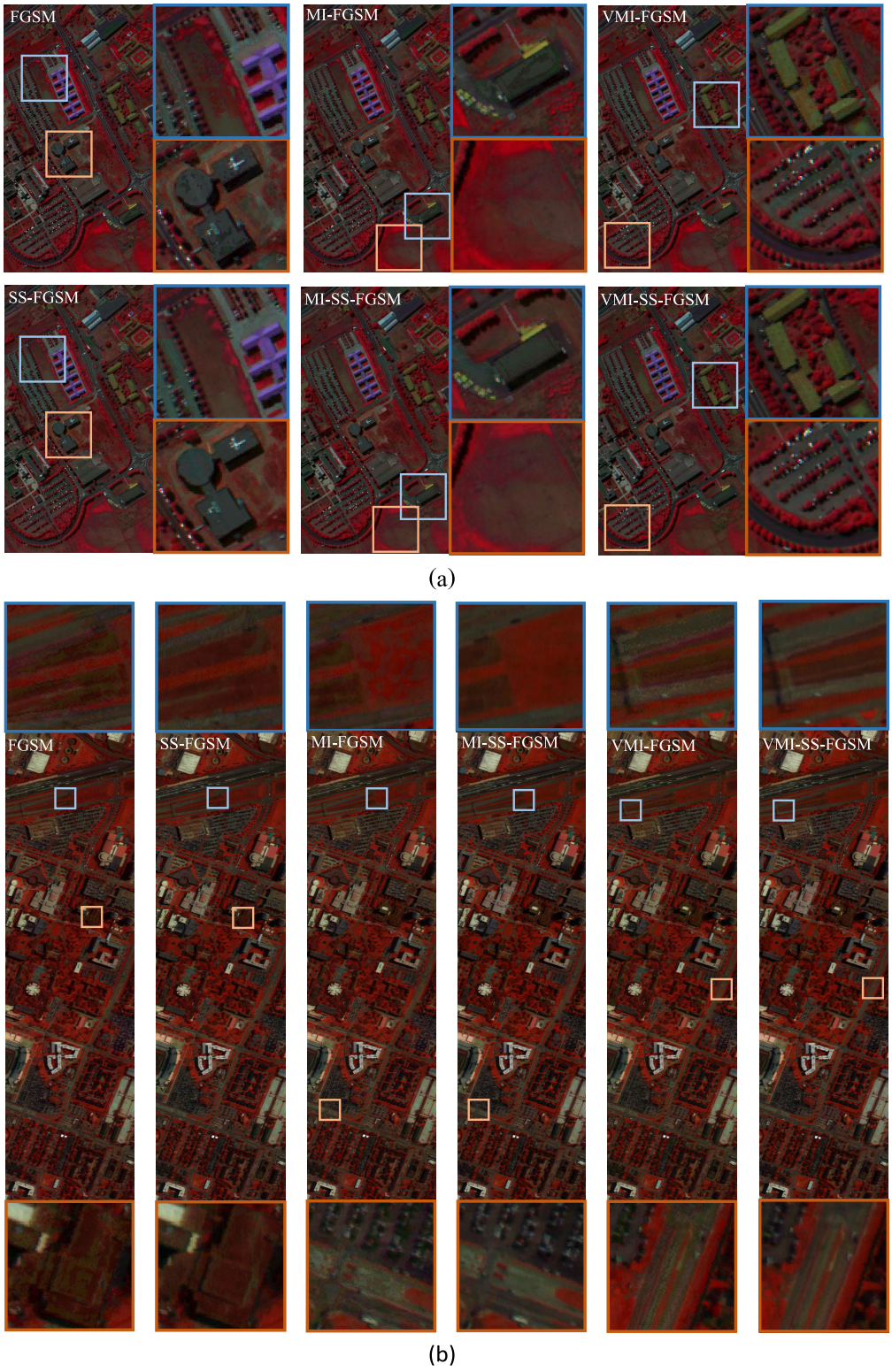


Fig. 8. Visualization of the generated adversarial examples. (a) PaviaU. (b) HoustonU 2018.

recommended to select an appropriate number of superpixels ranging from 4 to 8. This range allows for a balance between attack ability and attack robustness, considering both the effectiveness of the attack and the resilience to spatial resizing.

3) Spectral Clustering Size: The influence of spectral clustering size on attack performance is analyzed. The clustering number varies from 10 to 50, and the adversarial robustness is tested before and after spectral filtering. The result is reported in Fig. 7(b). The larger-scale clustering number means that each subclass has fewer spectral pixels. Therefore, before spectral filtering, the adversarial examples obtained by large-scale clustering achieve stronger attack ability, while the

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classification accuracy shows a trend of first decreasing and then increasing after spectral filtering. Hence, an appropriate spectral clustering size can be selected from 20 to 30 for the PaviaU dataset. Additionally, different HSI datasets contain different spectral bands; therefore, the spectral clustering size needs to be determined experimentally based on data characteristics.

E. Visual Results

In Fig. 8, we provide visualizations of the false color adversarial examples generated by both the baseline attack methods and our proposed methods. Comparing the adversarial examples of the baseline attack methods with those obtained by our method, it can be observed that the perturbations generated by our method exhibit a smoother pattern. This indicates that our method effectively promotes perturbation smoothing, resulting in perturbations that are less visually noticeable and have a more coherent and consistent appearance.

V. Conclusion

In this article, we proposed a universal object-level adversarial attack method in HSI classification. By addressing the limitations of existing methods, such as poor transferability and vulnerability to image-processing-based defenses, our proposed method introduces spatial and spectral consistency constraints to achieve smoother and more robust perturbations. The proposed method can be combined with existing attack methods to further improve their transferability and robustness. Experimental results on two real HSI datasets demonstrate the effectiveness of our method in improving transferability and robustness compared to state-of-the-art approaches. The proposed method has the potential to enhance the understanding of adversarial vulnerabilities in HSI classification and contribute to the development of more robust defense strategies in this field.

References

[1] Y. Duan, F. Luo, M. Fu, Y. Niu, and X. Gong, "Classification via structure-preserved hypergraph convolution network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507113.

[2] Z. Lv, P. Zhong, W. Wang, Z. You, and N. Falco, "Multiscale attention network guided with change gradient image for land cover change detection using remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[3] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713.

[4] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512913.

[5] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[6] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[7] S. Chen, Z. He, C. Sun, J. Yang, and X. Huang, "Universal adversarial attack on attention and the resulting dataset DAmageNet," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2188–2197, Apr. 2022.

[8] C. Luo, Q. Lin, W. Xie, B. Wu, J. Xie, and L. Shen, "Frequency-driven imperceptible adversarial attack on semantic similarity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15294–15303.

[9] X. Dong et al., "Robust superpixel-guided attentional adversarial attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12892–12901.

[10] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[11] H. Chen, "A hybrid clustering method based on k-means algorithm," in *Proc. 6th Int. Workshop Adv. Algorithms Control Eng.*, Oct. 2022, p. 10.

[12] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2014, pp. 1–11.

[13] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9185–9193.

[14] M. Li, C. Deng, T. Li, J. Yan, X. Gao, and H. Huang, "Towards transferable targeted attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 638–646.

[15] T. Li, M. Li, Y. Yang, and C. Deng, "Frequency domain regularization for iterative adversarial attacks," *Pattern Recognit.*, vol. 134, Feb. 2023, Art. no. 109075.

[16] J. Lin, C. Song, K. He, L. Wang, and J. E. Hopcroft, "Nesterov accelerated gradient and scale invariance for adversarial attacks," in *Proc. 8th Int. Conf. Learn. Represent.*, 2019, pp. 1–12.

[17] X. Wang and K. He, "Enhancing the transferability of adversarial attacks through variance tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1924–1933.

[18] C. Li, W. Yao, H. Wang, and T. Jiang, "Adaptive momentum variance for attention-guided sparse adversarial attacks," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 108979.

[19] C. Xie et al., "Improving transferability of adversarial examples with input diversity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2725–2734.

[20] Y. Dong, T. Pang, H. Su, and J. Zhu, "Evading defenses to transferable adversarial examples by translation-invariant attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4307–4316.

[21] C. Shi et al., "Multifeature collaborative adversarial attack in multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art no. 5631815, doi: 10.1109/TGRS.2022.3208337.

[22] M. Naseer, S. Khan, M. Hayat, F. S. Khan, and F. Porikli, "A self-supervised approach for adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 259–268.

[23] C. Shi, Y. Dang, L. Fang, Z. Lv, and M. Zhao, "Hyperspectral image classification with adversarial attack," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, Art no. 5510305, doi: 10.1109/LGRS.2021.3122170.

[24] C. Guo, M. Rana, M. Cissé, and L. V. D. Maaten, "Countering adversarial images using input transformations," in *Proc. 6th Int. Conf. Learn. Represent.*, 2018, pp. 1–12.

[25] F. Wang, G. Chai, Q. Li, and C. Wang, "MagNet with randomization defense against adversarial examples," in *Proc. Int. Conf. Netw. Netw. Appl. (NaNA)*, Dec. 2020, pp. 316–320.

[26] X. Yu, X. Wu, C. Luo, and P. Ren, "Deep learning in remote sensing scene classification: A data augmentation enhanced convolutional neural network framework," *GIScience Remote Sens.*, vol. 54, no. 5, pp. 741–758, Sep. 2017.

[27] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.

[28] J. Li et al., "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.

[29] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2015, *arXiv:1412.6572*.

[30] X. Tang, P. Yin, Z. Zhou, and D. Huang, "Adversarial perturbation elimination with GAN based defense in continuous-variable quantum key distribution systems," *Electronics*, vol. 12, no. 11, p. 2437, May 2023.

[31] P. Samangouei, M. Kabkab, and R. Chellappa, "Defense-GAN: Protecting classifiers against adversarial attacks using generative models," 2018, *arXiv:1805.06605*.

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[32] C. Shi, Y. Liu, M. Zhao, C.-M. Pun, and Q. Miao, "Attack-invariant attention feature for adversarial defense in hyperspectral image classification," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109955.
[33] F. Liao, M. Liang, Y. Dong, T. Pang, X. Hu, and J. Zhu, "Defense against adversarial attacks using high-level representation guided denoiser," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1778–1787.
[34] G. Jin, S. Shen, D. Zhang, F. Dai, and Y. Zhang, "APE-GAN: Adversarial perturbation elimination with GAN," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 3842–3846.
[35] C. Xie, W. Jianyu, Z. Zhishuai, R. Zhou, and Y. Alan, "Mitigating adversarial effects through randomization," in *Proc. ICLR*, 2017, pp. 1–16.
[36] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.
[37] L. Chen, Z. Xu, Q. Li, J. Peng, S. Wang, and H. Li, "An empirical study of adversarial examples on remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7419–7433, Sep. 2021.
[38] H. Li et al., "Adversarial examples for CNN-based SAR image classification: An experience study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1333–1347, 2021.
[39] C. Du, C. Huo, L. Zhang, B. Chen, and Y. Yuan, "Fast C&W: A fast adversarial attack algorithm to fool SAR target recognition with deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[40] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605111.
[41] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.
[42] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.
[43] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.
[44] C. Szegedy, S. Ioffe, and V. Vanhoucke, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 4278–4284.



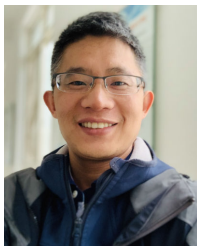
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